# 過去問

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## 1

(1) $x''-6x'+8x=0$

$\lambda^2-4\lambda+3=0$ とおく

$(\lambda-3)(\lambda-1)=0$

$\therefore \lambda=3, 1$

$\therefore x=C_1e^{3t}+C_2e^{t}$

(2) $x''-6x'+9x=0$

$\lambda^2-6\lambda+9=0$ とおく

$(\lambda-3)^2=0$

$\therefore \lambda=-3$ (2重解)

$x=e^{3t}(c_1+c_2t)$

(3) $x''-6x'+11x=0$

$\lambda^2-6\lambda+11=0$ とおく

$\lambda=\frac{6\pm\sqrt{36-44}}{2}=3\pm\sqrt{2}i$

$\therefore x=e^{3t}(c_1\cos\sqrt{2}t+c_2\sin\sqrt{2}t)$

(4) $x''-3x'=0$

$\lambda^2-3\lambda=0$

$\lambda(\lambda-3)=0$

$\lambda=3, 0$

$\therefore x=C_1e^{3t}+C_2$

(5) $x''-3x=0$

$\lambda^2-3=0$

$\lambda^2=3$

$\lambda=\pm\sqrt{3}$

$\therefore x=C_1e^{\sqrt{3}t}+C_2e^{-\sqrt{3}t}$

(6) $x''+3x=0$

$\lambda^2+3=0$

$\lambda^2=-3$

$\lambda=\sqrt{3}$

$=\sqrt{3}i$

$x=C_1\cos\sqrt{3}t+C_2\sin\sqrt{3}t$

## 2

(1) $x''-6x'+8x=8t^2+4t+14$

$x''-6x'+8x=0$ とおく

$\lambda^2-6\lambda+8=0$ とおく

$(\lambda-2)(\lambda-4)=0$

$\lambda=2, 4$

$\therefore x_0=C_1e^{2t}+C_2e^{4t}$ …①

$x_1=At^2+Bt+C$ と予想

$x_1'=2At+B$, $x_1''=2A$

$\therefore 2A-6(2At+B)+8(At^2+Bt+C)$

$=8t^2+4t+14$

$2A-12At-6B+8At^2+8Bt+8C$

$=8At^2+t(8B-12A)+2A-6B+8C$

$\begin{cases}8A=8\\8B-12A=4\\2A-6B+8C=14\end{cases}$ $\begin{cases}A=1\\B=2\\C=3\end{cases}$

$x_1=t^2+2t+3$ …②

①と②より

$x=x_0+x_1$

$=C_1e^{2t}+C_2e^{4t}+t^2+2t+3$

(2) $x''-6x'+8x=-4e^{3t}$

$x_0''-6x_0'+8x_0=0$ とおく

$\lambda^2-6\lambda+8=0$ とおく

$(\lambda-4)(\lambda-2)=0$

$\therefore \lambda=4, 2$ $\therefore x_0=C_1e^{4t}+C_2e^{2t}$ …①

$x_1=Ae^{3t}$ と予想

$x_1'=3Ae^{3t}$, $x_1''=9Ae^{3t}$

$\therefore 9Ae^{3t}-18Ae^{3t}+8Ae^{3t}=-4e^{3t}$

$e^{3t}(9A-18A+8A)=-4$

$-Ae^{3t}=-4$

$A=4$

$\therefore x_1=4e^{3t}$ …②

$\therefore$ ①②より

$x=x_0+x_1$

$=C_1e^{4t}+C_2e^{2t}+4e^{3t}$

(3) $x''-6x'+8x=-10e^{2t}$

$x''-6x'+8=0$ とおく

$\lambda^2-6\lambda+8=0$ とおく

$x_0=C_1e^{2t}+C_2e^{4t}$ …①

$x_1=Ate^{2t}$

$x_1'=Ae^{2t}+2Ate^{2t}$

$x_1''=2Ae^{2t}+2Ae^{2t}+4Ate^{2t}$

$\therefore 4Ae^{2t}+\cancel{4Ate^{2t}}-6Ae^{2t}-\cancel{12Ate^{2t}}+\cancel{8Ate^{2t}}=-10e^{2t}$

$-2Ae^{2t}=-10e^{2t}$ $\therefore A=5$

$\therefore x_1=5te^{2t}$

$\therefore x=x_0+x_1=C_1e^{2t}+C_2e^{4t}+5te^{2t}$

(4) $x''-6x+8x=10\cos 2t$

$x_0=C_1e^{2t}+C_2e^{4t}$

$x_1=A\cos 8t+B\sin 8t$ とおく

$x_1'=-A8\sin 8t+B8\cos 8t$

$x_1''=-A8^2\cos 8t-B8^2\sin 8t$

$\therefore -6x'=-6A8\sin 8t-6B8\cos 8t$

$8x=8A\cos 8t+8B\sin 8t$

$(-8B8^2-6A8+8B)\sin 8t+(-A8^2-6B8-8A)\cos 8t$

$=10\cos 2t$

$\begin{cases}-B8^2-6A8+8B=0\\-A8^2-6B8-8A=10\end{cases}$ $\begin{cases}-4B-12A+8B=0\\-4A-12B-8A=10\end{cases}$

$\therefore \begin{cases}-16A+8B=0\\-12A-12B=10\end{cases}$ $\therefore \begin{cases}8B=16A \therefore B=2A\\36A=10 \therefore A=\frac{10}{36}=\frac{5}{18}\end{cases}$

$B=\frac{5}{9}$

$x=x_1+x_0=C_1e^{2t}+C_2e^{4t}+\frac{5}{18}\cos 2t+\frac{5}{9}\sin 2t$

## 3

(1) $x''=t+2+e^{3t}+\cos 4t$

$x'=\int x''\,dt=\frac{1}{2}t^2+2t+\frac{1}{3}e^{3t}+\frac{1}{4}\sin 4t$

$x=\int x'\,dt=\frac{1}{6}t^3+t^2+\frac{1}{9}e^{3t}-\frac{1}{16}\cos 4t$

(2) $t^2x''-6tx'+12x=0$

$x=t^\lambda$, $x'=\lambda t^{\lambda-1}$

$x''=\lambda(\lambda-1)t^{\lambda-2}$

$t^2(\lambda^2-\lambda)t^{\lambda-2}-6t(\lambda t^{\lambda-1})+12t^\lambda=0$

$\lambda^2t^\lambda-\lambda t^\lambda-6\lambda t^\lambda+12t^\lambda=0$

$\lambda^2t^\lambda-7\lambda t^\lambda+12t^\lambda=0$

$(\lambda^2+5)t^\lambda=0$

$\lambda^2=-5$ $\therefore \lambda=\sqrt{-5}$ $\therefore \lambda=\sqrt{5}i$

$\therefore x=C_1\cos\sqrt{5}t+C_2\sin\sqrt{5}t$

(3) $\begin{cases}x'=4x-6y-12t+2 \cdots ①\\y'=x-3y \cdots ②\end{cases}$

$x=3y+y'$ $\therefore x'=3y'+y''$

①に代入

$3y'+y''=4x-6y-12t+2$

$=4(3y+y')-6y-12t+2$

$y''-y'-6y=-12t+2$ …③

$y''-y'-6y=0$ とする

$\lambda^2-\lambda-6=0$

$(\lambda+2)(\lambda-3)=0$

$\lambda=-2, 3$

$y_0=C_1e^{-2t}+C_2e^{3t}$

$y_1=At^2+Bt+C$ と予想

$y_1'=2At+B$, $y''=2A$

$\therefore 2A-2At-B-6At^2-6Bt-6C=-12t+2$

$-6At^2+(-2A-6B)t+2A-6C-B=-12t+2$

$\begin{cases}-6A=0\\-2A-6B=-12\\2A-6C-B=2\end{cases}$ $\therefore \begin{cases}A=0\\B=2\\C=\frac{4}{6}=-\frac{2}{3}\end{cases}$

(4) $x''=\frac{2tx'}{t^2+1}$

$x'=p$ より

$x'=p'=\frac{dp}{dt}$

$\therefore \frac{dp}{dt}=\frac{2tp}{t^2+1}$

$\int\frac{1}{p}dp=\int\frac{2t}{t^2+1}dt$

$\log|p|=\log|t^2+1|+C$

$=\log|t^2+1|+\log e^C$

$p=(t^2+1)\cdot\pm e^C$

$=(t^2+1)c$

$x'=(t^2+1)c$

$\int x'\,dt=c\int t^2+1\,dt$

$=c(\frac{1}{3}t^3+t)+A$

## 4 応用

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